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$$\left(\frac{\sqrt{5} + \sqrt{15}i}{5}\right)^6$$
$$r = \sqrt{\frac{5}{25} + \frac{15}{25}} = \frac{2}{\sqrt{5}}$$

$$\sin \theta = \frac{\frac{\sqrt{15}}{5}}{\frac{\sqrt{\frac{4}{5}}}{\sqrt{5}}} = \frac{\sqrt{3}}{2}$$

$$\cos \theta = \frac{\frac{\sqrt{5}}{5}}{\frac{\sqrt{\frac{4}{5}}}{\sqrt{5}}} = \frac{1}{2}$$

Eerste kwadrant:

$$\theta = \frac{\pi}{3}$$

$$z = \left(\frac{2}{\sqrt{5}}\right)^6 (\cos(2\pi) + i \sin(2\pi))$$

$$z = \frac{64}{125}$$

References